ANALYTICS CASE STUDY

Customer shopping behavior, decoded

Turn 3,900 transactions into a practical analytics case study: how the data was prepared, what it revealed, and where the business can act.



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01 · DATASET

The dataset is broad enough to connect behavior to business outcomes

3,900 purchases

18 columns spanning demographics, products, promotions, shipping, ratings, and subscriptions.

Four product categories

Accessories, Clothing, Footwear, and Outerwear.

Core outcome measures

Purchase amount, revenue, review rating, repeat behavior, and subscription status.

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Data preparation turned raw transactions into analysis-ready signals

The workflow moved from inspection to cleaning, feature creation, and SQL analysis.

01

Inspect

Checked structure and descriptive statistics in Python

02

Impute

Filled 37 missing review ratings with category-level medians

03

Shape

Standardized column names, added age groups and purchase-frequency days, and removed redundant promo-code data

04

Load

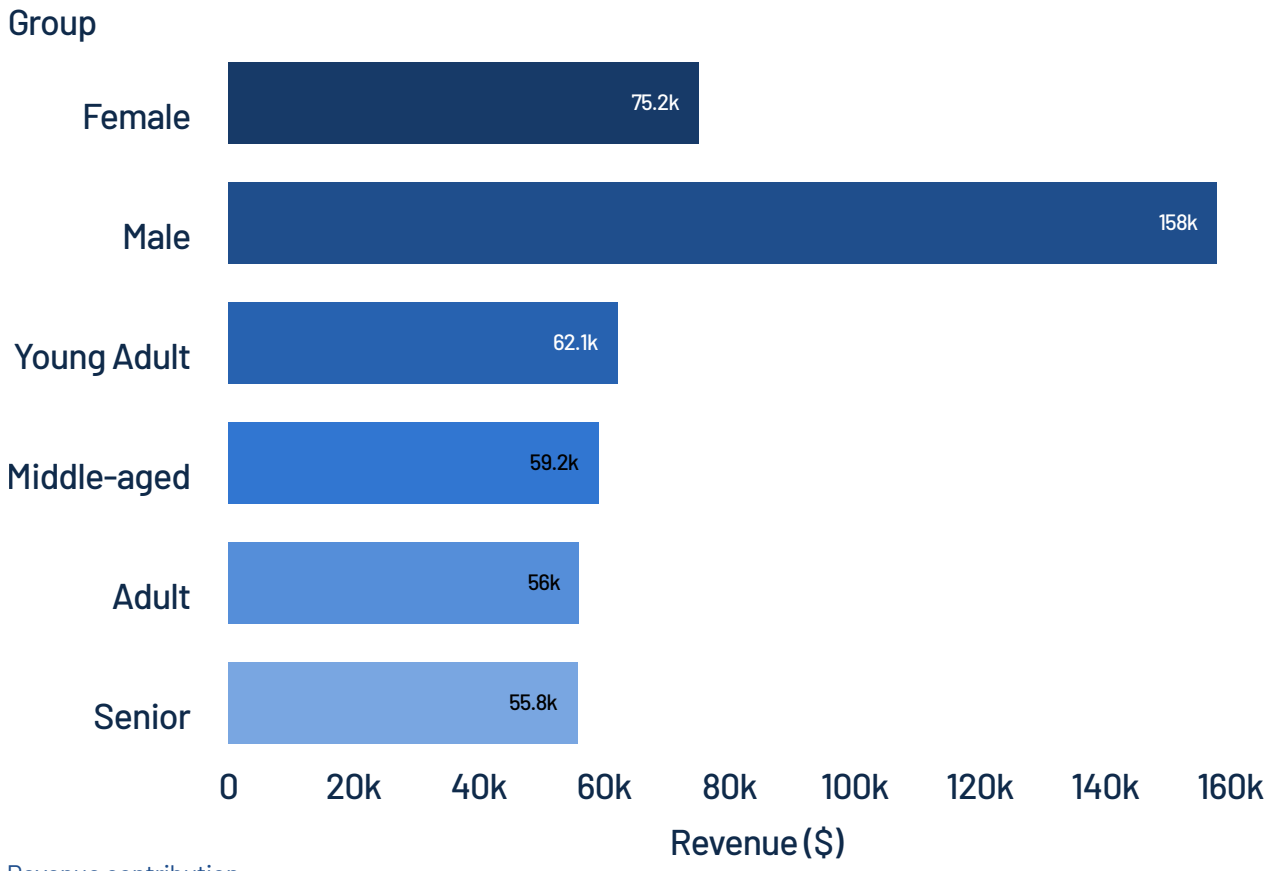
Loaded the cleaned DataFrame into PostgreSQL for SQL analysis

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Revenue is concentrated in male customers and younger cohorts

Male customers generated **$157,890** versus **$75,191** from female customers.

Young Adults led the age groups at **$62,143**, followed by Middle-aged at **$59,197**, Adults at **$55,978**, and Seniors at **$55,763**.



Revenue contribution

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04 · ASSORTMENT

Clothing is the commercial center of the assortment

Clothing leads revenue, while product leaders and ratings reveal where the assortment is strongest.

Clothing

$100K

Blouse, Pants, and Shirt lead the category.

Accessories

$75K

Jewelry, Sunglasses, and Belt lead the category.

Footwear

$35K

Sandals, Shoes, and Sneakers lead the category.

Outerwear

$20K

Jacket and Coat lead the category.

Top-rated products: Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat 3.80 · Skirt 3.78

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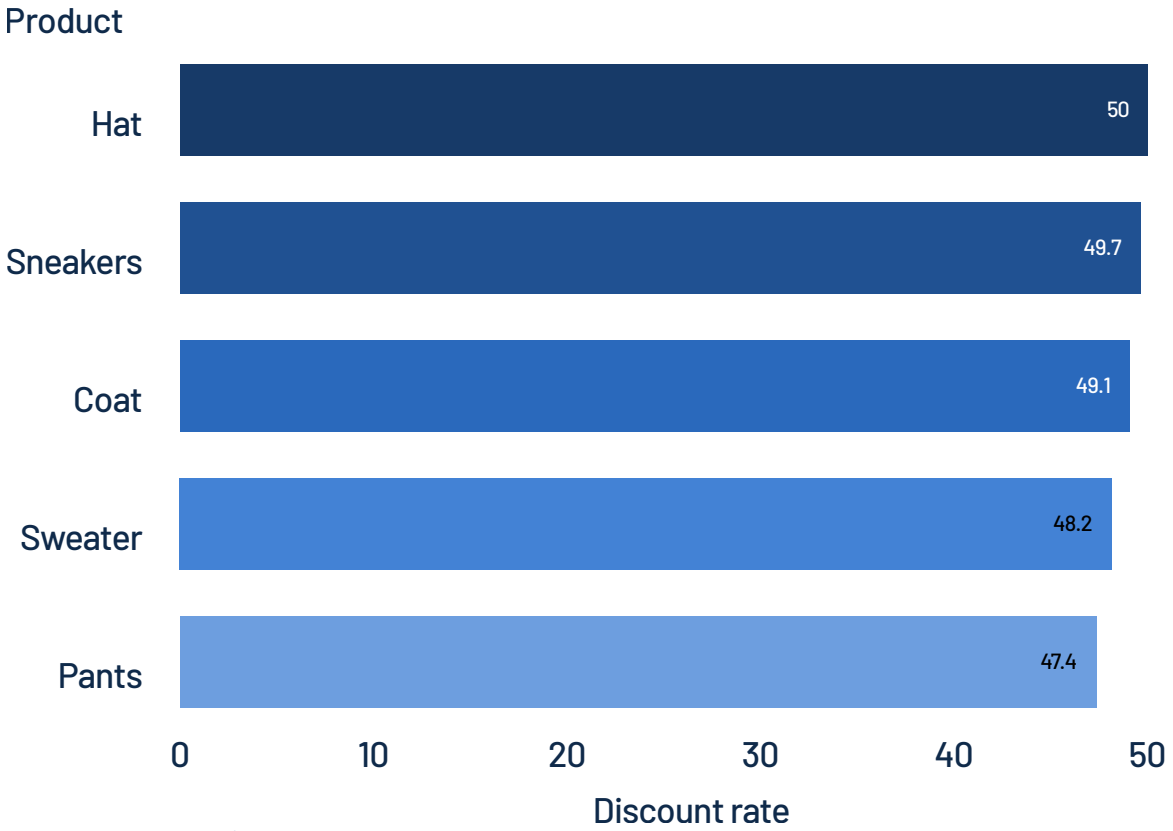
05 · PROMOTION

Discounts attract high spend, but dependence is uneven

839 customers used discounts while spending above the average purchase amount.

Discounting can lift conversion, but the product mix suggests a need to protect margin where discounting is already the norm.

Most discount-dependent products



Share of purchases discounted

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06 · SHIPPING

Shipping speed edges up average order value

STANDARD SHIPPING

$58.46

Average purchase amount

EXPRESS SHIPPING

$60.48

Average purchase amount

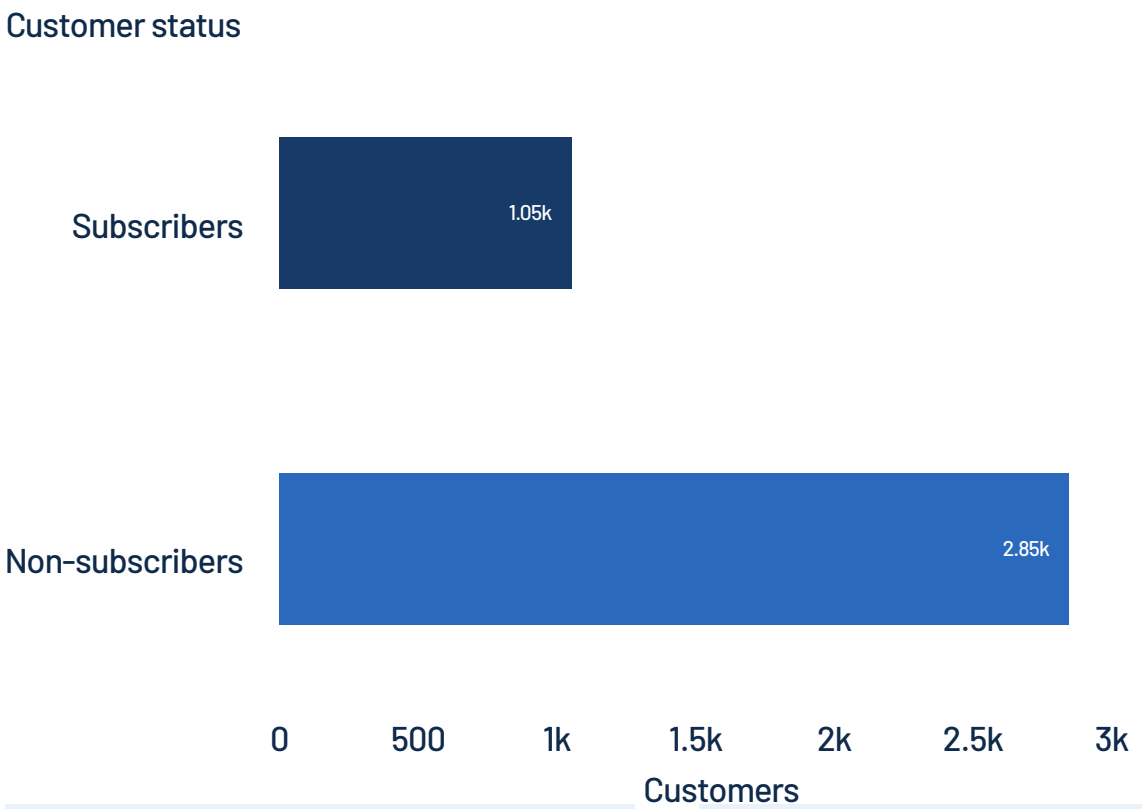
The gap is modest, but faster delivery may support a higher-value basket.

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Subscriptions are underpenetrated, not underperforming

Subscribers contributed **$62,645**, while non-subscribers contributed **$170,436**.

The opportunity is adoption and retention, not a clear average-spend premium.



SUBSCRIBER AVERAGE

$59.49

NON-SUBSCRIBER AVERAGE

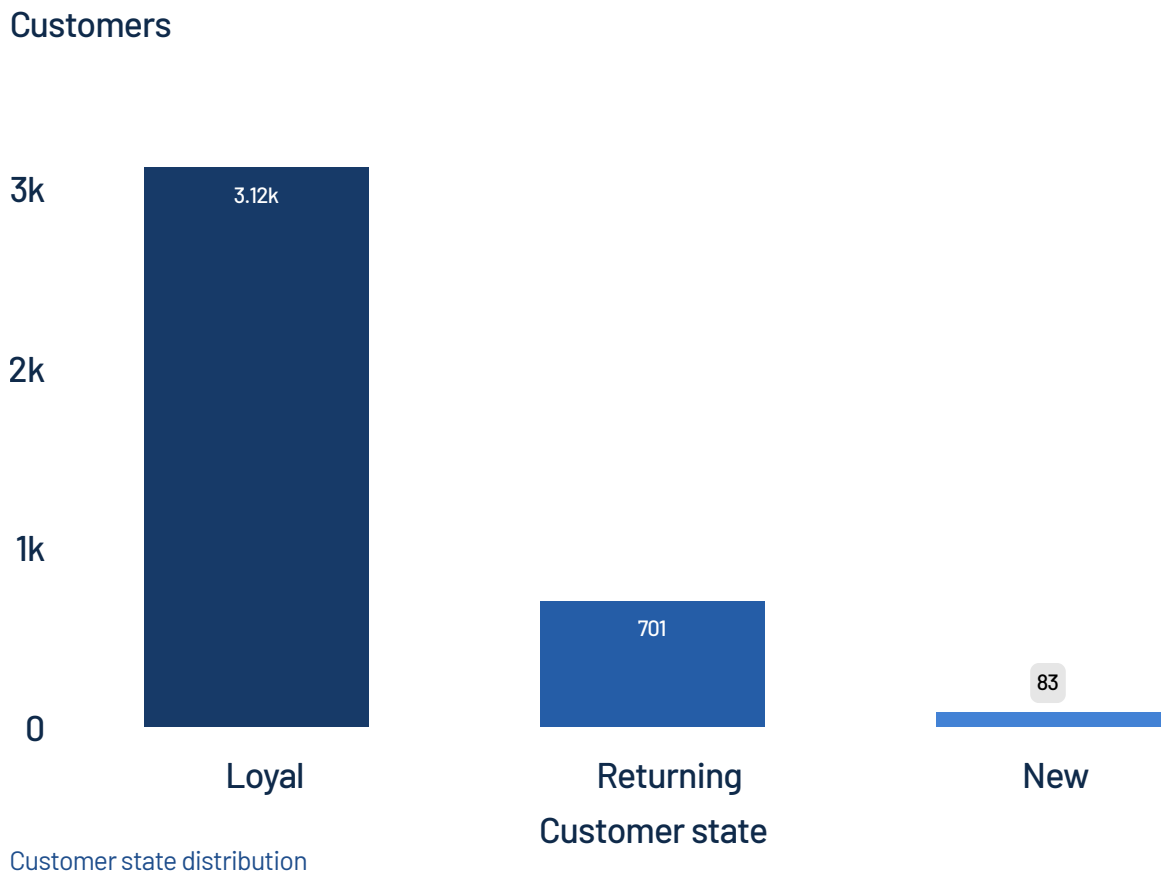
$59.87

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Loyalty is already the dominant customer state

3,116 customers are Loyal, compared with **701** Returning and **83** New customers.

Among repeat buyers, **958** have subscriptions versus **2,518** without subscriptions.



Customer state distribution

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The Power BI dashboard turns analysis into a reusable decision surface

Headline metrics: 3.9K customers, $59.76 average purchase amount, and 3.75 average review rating.

Interactive cuts: subscription status, gender, category, shipping type, and age group.

Visual outputs: revenue and sales by category and age group.



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